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Palaeomagnetism and age of the quartz-porphyry intrusions, Isle of Arran

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SYNOPSIS

Most of the quartz-porphyry intrusions of Arran, many of which form the acid member of composite intrusions, have been sampled palaeomagnetically. Though both normal and reversed polarity may be found in a single intrusion it is concluded that the reversed polarities are due to overprinting by later reversely magnetised intrusions and that all the quartz-porphyries were intruded while the magnetic field had normal polarity.

The age of the Drumadoon quartz-porphyry dyke and, by inference, that of all the other quartz porphyry intrusions, is $58{\cdot}5 \pm 0{\cdot}8$ (1σ) Ma.

It is suggested that normal polarity may provide a stratigraphic 'horizon' useful for elucidating the poorly-known succession of Arran Tertiary igenous rocks, most of which are reversely magnetised.

This work forms part of a comprehensive study, using palaeomagnetic polarity and radiometric dating, of the timing and duration of activity in the British Tertiary Igneous Province, and the age quoted above lies in the period of peak activity. This is the first time that a correlation of polarity with a distinctive petrographic rock type has been found.

INTRODUCTION

The Tertiary igneous succession of Arran is poorly known because of a lack of suitable cutting relationships between the various igneous masses, and attempts at correlation using petrographic criteria have failed because of the recurrence of magma types. Palaeomagnetic polarity may provide a stratigraphic constraint so the quartz-porphyries, which form a distinctive group, were subjected to palaeomagnetic and isotopic dating studies.

The quartz-porphyries of Arran occur as dykes, sills and a plug (Fig. 1). Several form part of a composite intrusion, that is two or more distinct magma types (in this case acid and basic) have been injected so closely in time that there is no chilling of one against the other, though sometimes it may be possible to determine the relative order of intrusion by the presence of e.g. inclusions of one type in the other.

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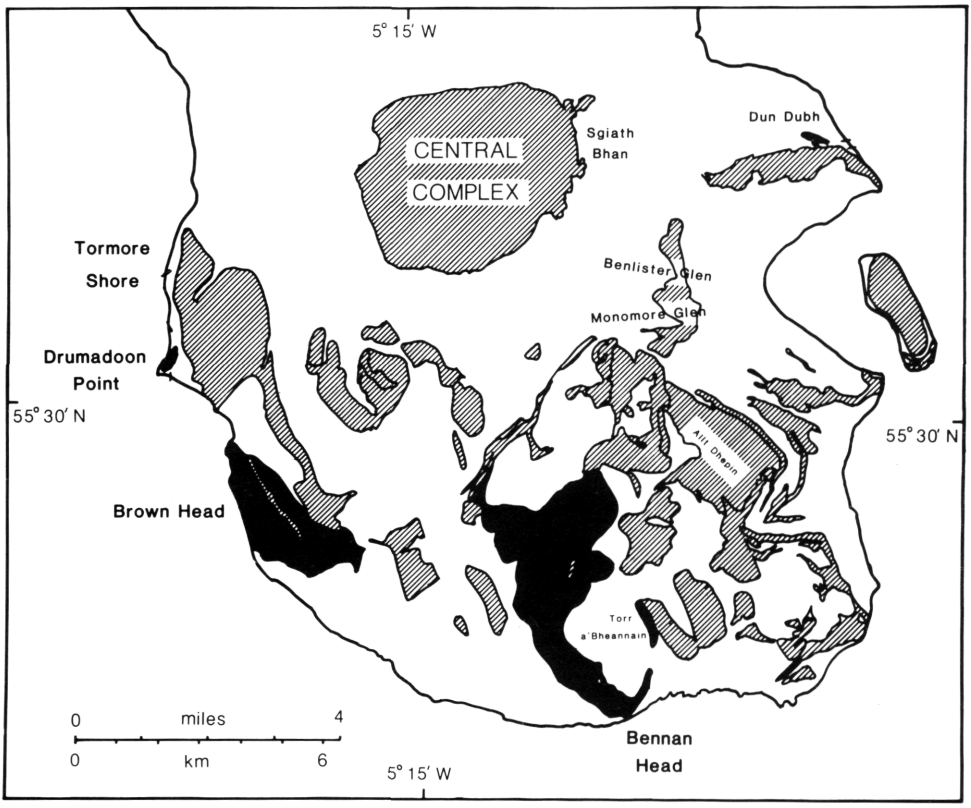


FIG. 1. Simplified geology map of southern Arran, showing quartz-porphyries in black, with other Tertiary igneous bodies hatched. (Based on the British Geological Survey, 1 inch:1 mile map).

Because the component parts of a composite intrusion have been injected within such a short space of time, they should have the same palaeomagnetic direction and certainly the same polarity. However, in the course of an investigation of the palaeomagnetic stratigraphy of the Arran Tertiary sequence (Dagley and Hodgson 1979; Hodgson 1981) both normal (N) and reversed (R) polarities were found in the Judd 4 and Cleiteadh nan Sgarbh composite dykes on the west coast, and this led to a more intensive survey of these bodies, in addition to measurements of other quartz-porphyries.

SAMPLING AND LABORATORY MEASUREMENTS

A hand-held diamond-tipped drill was used to cut cores which were oriented non-magnetically before being detached. These cores, 2·5 cm in diameter, were cut into specimens 2·5 cm long which were used in the magnetic measurements.

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TABLE 1

Palaeomagnetic data

Judd 4 Dyke

	N	R	D	I	α_{95}	Polarity
Quartz Porphyry						
402(6)			10·9	66·0		N
411B(6–26)	16	6·86	22·6	69·4	37·9	N
415A(1–2)	4	3·94	254·2	−76·3	13·6	R
415B(3–4)	4	4·00	8·14	58·7	3·0	N
427	6	6·00	7·86	63·1	0·8	N
401	6	6·00	357·5	64·1	1·9	N
413B(6, 7, 8)	6	6·00	163·2	−73·3	1·5	R
422B(3–9)	7	6·80	170·3	−61·3	11·3	R
422C(12–14)	6	6·00	168·9	−72·7	1·3	R
422D(16–18)	6	6·00	8·6	60·4	1·1	N
Means	N		5·7	61·7	4·0	N
($\alpha_{95} < 10°$)	R		166·1	−73·0	3·9	R
N.W. Margin						
Body 3						
400	8	8·00	13·6	53·7	3·9	N
Body 7						
411A(1–5)	12	11·61	42·8	60·5	8·3	N
423A(1–2)	4	4·00	39·0	60·0	4·3	N
423B(3–4)	4	3·99	164·8	−70·9	6·2	R
402(1–5)	6	5·79	343·2	68·1	14·1	N
405	8	7·95	2·4	57·0	4·8	N
Mean (excluding 423) ($\alpha_{95} < 10°$)			23·5	58·9	12·2	N
S.E. Margin						
Body 5						
403(1–17)	15	7·34	26·6	50·5	41·3	N
403(4, 6, 11, 12, 13, 16) Clusters	6	5·85	20·6	67·6	11·7	N
403A(5, 6, 8)	6	5·77	63·3	61·6	14·7	N
403B(12–15)	8	7·76	345·3	60·2	10·5	N
403C(18, 19)	4	4·00	173·8	−74·1	1·1	R
422(10)	2		5·3	22·4		I
Body 6						
404	8	8·00	185·2	−74·0	1·9	R
426	6	6·00	177·2	−74·0	1·0	R
Body 9						
420	4	4·00	13·4	55·6	0·7	N
416	6	6·00	6·4	68·0	1·4	N
Means	N (416 + 420)		10·6	61·8	28·4	N
($\alpha_{95} < 10°$)	R (403C, 414, 426)		178·7	−74·1	2·4	R
Basaltic Dyke						
Body 4						
414	8	7·00	145·4	−68·3	22·4	

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TABLE 1 (contd.)

Judd 4 Dyke (contd.)

	N	R	D	I	α_{95}	Polarity
Basaltic Dyke (contd.)						
Body 8						
421	6	5·97	41·0	76·2	5·2	N
411C(27–30)	8	6·10	322·4	−58·0	33·5	I
Amygdaloidal Dolerite Dykes						
Body 2						
412	8	8·00	186·3	−65·5	0·8	R
422A(1, 2, 11)	6	5·99	175·7	−67·7	2·4	R
Cliff Site						
417	6	5·48	169·5	−82·5	23·0	R
Mean (excluding 417) ($\alpha_{95} < 10°$)			181·2	−66·7	10·3	R
Thin Basic Dyke						
Body 1						
413A(1–5)	8	8·00	177·9	−71·1	1·3	R
Body 1A						
425	6	6·00	168·2	−78·1	0·7	R
Mean			174·1	−74·7	16·2	R
Olivine Dolerite						
Body 10						
Hodgson 115	8	7·98	162·6	−63·2	2·0	R
Sheet						
424	6	5·60	9·7	−73·7	19·6	

Cleiteadh Nan Sgarbh Dyke

Quartz Porphyry						
408B(5–14)	18	17·96	194·7	−76·7	1·6	R
408C(15–17)	6	5·99	14·2	72·8	2·5	N
409B(8)	—	—	304·5	73·7	—	N
410A(4)	—	—	44·7	1·2	—	I
410A(2–4)	—	—	47·0	43·7	19·9	N
105	10		6	63	4·2	N
107	4		4	77	4·7	N
Margin						
SW 409A(1–6)	12	11·99	12·3	61·0	1·6	N
NE 410B(8)	—	—	175·8	−66·6	—	R
410B(5, 6, 8)			174·9	−68·2	8·5	R
283	10		186	−65	1·2	R
Felsite						
410C(9, 10, 12)	6	5·99	186·4	−71·1	2·93	R
108	4		158	−65	7·4	R

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TABLE 1 (contd.)

	N	R	D	I	α_{95}	Polarity
Cleiteadh Nan Sgarbh Dyke (contd.)						
Dykes						
408A(1–4)	8	8·00	187·2	−78·4	1·2	R
106	6	5·99	221·3	−73·1	1·3	R
Drumadoon Sill						
Quartz Porphyry						
221	12		25	65	2·7	N
103	12		27	63	6·7	N
Mean			26	64	4·8	N
Margins						
218	12		13	64	1·8	N
219	12		23	55	3·2	N
Mean			18·7	59·6	22·7	N
Drumadoon Dyke						
Quartz Porphyry						
117	8		15	69	1·1	N
Brown Head Sill						
Quartz Porphyry						
118	12	11·82	2·9	65·7	9·0	N
246	12	11·99	5·5	11·9	1·4	I
Q003	—	—	350·3	72·6	4·1	N
Bennan Head Sill						
Quartz Porphyry						
150	8		338	66	10·1	N
Q004	—	—	3	72·6	1·4	N
Margins						
151	12		0·9	67	1·2	N
200	12		1·8	59	1·5	N
Mean			1·4	63	17·5	N

Specimens were progressively demagnetised by alternating magnetic fields to find a stable direction of magnetisation, believed to have been acquired when the intrusion cooled, and the results for several independently oriented specimens from the same body were used to define a mean direction and error (at the 95% level of confidence, expressed as a cone of semi-angle α_{95}). In addition, a few specimens were thermally demagnetised. In those cases where a.f. demagnetisation of one specimen could be compared with thermal demagnetisation of a second specimen from the same core, the thermal cleaning generally gave less well defined directions, though agreeing with the directions obtained using alternating fields,

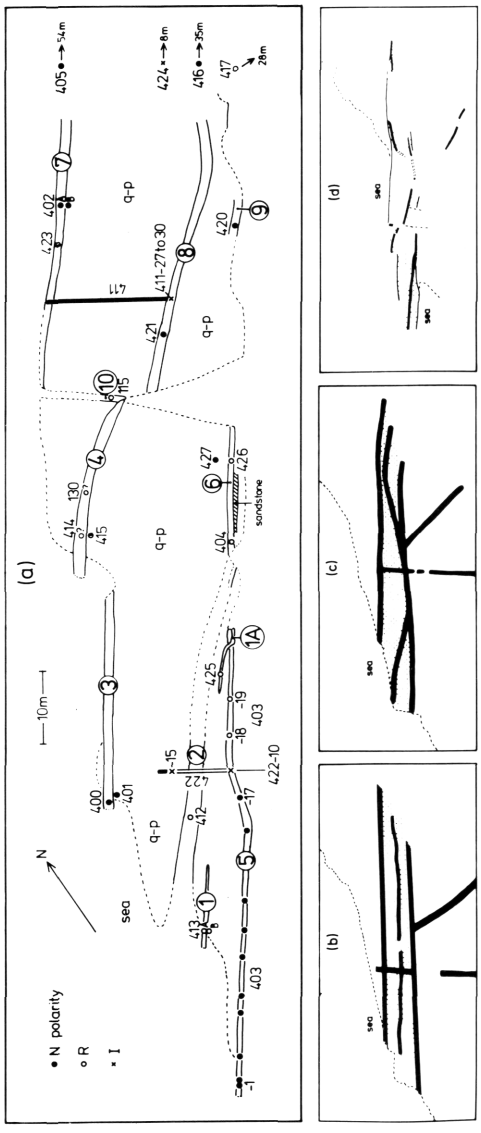


FIG. 2. Judd 4 dyke. (a) shows polarities measured in various parts of the composite and intersecting dykes. The bulk of the Judd 4 dyke is of quartz-porphyry (q-p), between narrow margins of dolerite: sheets 3, 5, 7 and 9. The remaining numbered sheets are later. See text for further details. (b), (c) and (d) are drawings by, respectively, Judd (1893), Tomkieiff (1969) and Kanaris-Sotiriou and Gibbs (1985), with quartz-porphyry shown dotted and basic margins and dykes in black.

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and contributed no new information. Further details of sampling and measuring methods may be found in Dagley *et al.* (1978).

Samping was carried out over several seasons, and sampling sites are shown in Figures 1, 2 and 4. Directions of magnetisation are given in Table 1 and shown for a traverse across Judd 4 on Figure 3, while Figure 2 shows polarities for sites in Judd 4.

Results

Judd 4

The dykes of the Tormore shore were studied by Judd (1893) and his numbering has been retained. They include one quartz-porphyry, the Judd 4 dyke. This is a composite dyke about 25 m wide, the bulk being of quartz porphyry but with tholeiitic dolerite margins about a metre wide. It is also cut by basic sheets, most of which are vertical. During the course of our work it was realised that the maps of Judd 4 given by Judd (1893) and by Tomkieiff (1969) (Fig. 2, b and c) are inaccurate and a new map was drawn up with the aid of a 200 m tape stretched alongside the dyke and profiles were measured across the dyke every 10 m, with additional measurements where required. The combination of low tides and absence of seaweed permitted a more extensive examination of the dyke than had previously been possible.

Our map (Fig. 2a) shows that the Judd 4 dyke is not only considerably more complicated than as shown by Judd or Tomkeieff but it differs from theirs on several points, notably that the margins cannot be traced straight across the cross-cutting dyke (body 10), and that the SW part of the dyke has been extensively eroded by the sea. Further, the apparent branch in the NE margin could not be seen, possibly because of poor exposure.

The palaeomagnetic polarities are shown on Figure 2 and at first sight fit no simple pattern. Both normal and reversed polarities are found in both doleritic margins (bodies 3, 5, 6, 7, 9), in the quartz-porphyry and in the intersecting sheets (bodies 4 and 8). However, the explanation seems to be that the composite dyke proper, viz. the quartz-porphyry plus dolerite margins, is normal but is cut by reversely magnetised basic sheets which have remagnetised some parts of it. This interpretation will now be justified.

Close inspection in the field showed that bodies 1 and 1a (thin basic dykes), 2 (amygdaloidal dolerite) and 4 and 8 (basalt dykes) chilled against the quartz porphyry and are therefore later. The cross-cutting olivine-dolerite dyke, body 10, was only just exposed during our visits, but enough was visible to show chilling against body 4, and so this must belong to a yet later generation. No chilling was observed between the marginal dolerites (bodies 3, 5, 6, 7, and 9) and

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the quartz-porphyry. These observations agree with the conclusions of Kanaris-Sotiriou and Gibbs (1985).

Petrological examination shows that the basic dykes 1 and 1a are probably related, as are the basaltic dykes 4 and 8. Therefore there are probably four distinct dykes or sheets cutting the composite body and at least three phases of intrusion.

Bodies 1, 1a, 2 and 10 are all clearly reversely magnetised. The magnetic results for bodies 4 and 8 are less clear as many of the samples proved to be unstable. At sites 414 and 130 the dyke is reversely magnetised though with large α_{95}. A poorly defined intermediate direction was obtained for site 411C while three cores at site 421 gave a mean direction which is normal. However, it was found that the quartz-porphyry in close contact with this dyke gives magnetically stable results; core 26 of traverse 411 (see Fig. 3a) and two cores from a group at site 415 give reversed directions. The balance of the evidence suggests that this dyke, too, is

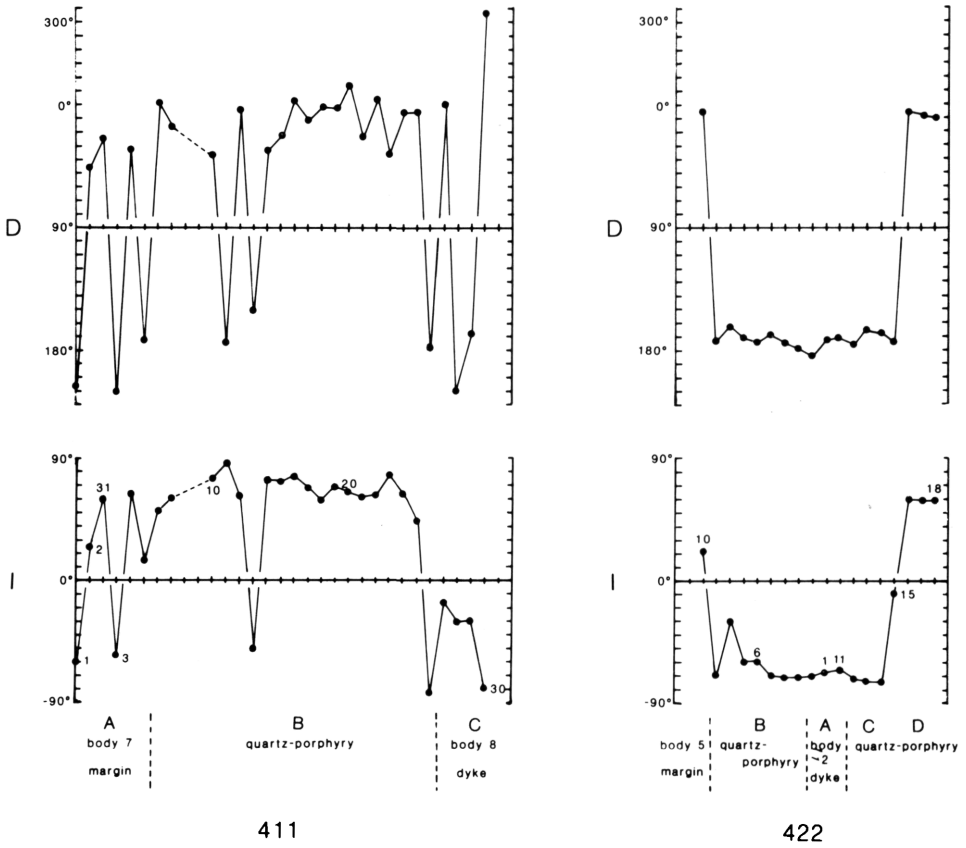


FIG. 3. Palaeomagnetic profiles across the Judd 4 dyke. Directions have been resolved into declination (D) and inclination (I) and plotted separately.

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reversely magnetised and therefore all of the bodies which cut the composite are reversely magnetised.

The sub-horizontal sheet sampled at site 424 gives an intermediate direction with a very steep negative dip and in view of the large α_{95} this, too, is interpreted to be reversely magnetised.

Generally, cores from the dolerite forming body 7 of the NW margin of the composite were relatively unstable. At site 423 two cores gave normal and two gave reversed directions, whilst only three of the five cores at site 402 nearby could be used to give a poorly defined normal direction, as did the cores from the traverse 411A. However, site 400 in body 3 to the south and site 405 to the north both gave a clear normal polarity and it is concluded the NW margin is normally magnetised.

Results from the SE doleritic margin are conflicting. Clusters of cores at sites 403A and 403B in body 5 give poorly defined normal directions but two cores at 403C are clearly reversed. Body 6 (sites 404 and 426) is clearly reversed but body 9 (site 420 and site 416) gives well defined normal mean directions.

Similarly, samples from the quartz-porphyry give normal mean directions at sites 401 and 427, and in traverses 411B and 422D, as does a single core adjacent to the NW margin at site 402, whereas well-defined reversely magnetised directions are found for site 413B, parts of traverses 411 (core 26) and 422 (parts B and C) and two cores at site 415.

These results can be explained if the intrusion of body 2 and possibly also bodies 1 and 1A have caused remagnetisation of the neighbouring rock to some distance on either side. From other work on the dykes of the British Tertiary Igneous Province complete overprinting does not usually extend to more than a half-dyke width (e.g. Brown 1981). Such a narrow overprinting width seems to be consistent with the R polarity of sample 411-26 close to internal dyke 8, but N polarity of samples further away (see Fig. 3). However, traverse 422 shows that the internal dyke 2 has overprinted to more than its width away, extending to the SE margin where 422-10 gives an intermediate direction (see Fig. 3); this may have occurred because a large volume of magma travelled up the fissure. This wide band of overprinting accounts for the apparently incomprehensible result that the SE margin is N at its southward end, but is R further north, the R polarities of 403-18 and -19 being explained because the margin and intruding dyke 2 are closer together here. The R polarity of sites 404 and 426 probably arise because of the proximity of dyke 2 which is here separated from the composite dyke only by a narrow screen of sandstone.

Bodies 1 and 1a are so close to body 2 that it is not possible to tell if their R polarity is primary or due to overprinting by body 2; however, as all the other intersecting dykes are R it is likely that it too has contributed to the overprinting of nearby quartz-porphyry and dolerite margin. Indeed, the unusually wide band of overprinting revealed by traverse 422 may be the result of two contributory

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intrusions in this region for, further north, where possibly only body 2 is effective in causing overprinting, site 426 in the margin has been overprinted but 427 in the quartz-porphyry remains normal.

The incongruent directions found in the NW margin and traverse 411 are not obviously explained in this way. One possibility is that they have been produced by a horizontal sheet, similar to 424 for example, which has since been eroded.

This interpretation, that the Judd 4 composite dyke was intruded during a normal interval, but has in places suffered partial overprinting, is supported by the observation that all the other sampled quartz-porphyry intrusions of Arran, where they are not in close proximity to later, reversely magnetised, basic minor intrusions also give N polarity.

The misalignment of bodies 4 and 8, which petrologically are similar, suggests a dextral offset along the line of the cross-cutting dyke 10. However, the misalignment of the margins 3 and 7 suggests the opposite sense of offset. It is considered likely that the alignment of internal dyke 4/8 merely results from its following a line of weakness closest to its preferred direction; dykes 1 and 2 show offsets of similar sense where they cross the SE margin.

As stated earlier, the apparent branch in the SE margin of the composite intrusion shown in the maps of Judd and Tomkieiff could not be observed. However, a dyke was seen in the cliffs, site 417, trending obliquely towards Judd 4, and as it is petrographically similar to internal dyke 2 and of R polarity (though with large α_{95}) it is thought to be dyke 2 which has swung away from the alignment of Judd 4 after trending parallel to it for an unknown distance. If so, it suggests that the large-scale trend of internal dyke 2 (and perhaps of the other internal dykes) is quite oblique to the trend of Judd 4, but it was constrained to intrude sub-parallel to Judd 4, presumably because the trend of Judd 4 was still a line of weakness.

Cleiteadh nan Sgarbh dyke

This composite dyke consists of a 15·5 m thick quartz-porphyry flanked by 0·6 m thick dolerite margins. It outcrops on the coast about $1\frac{1}{4}$ km to the south of the Judd 4 dyke, dipping steeply to the NE. Although it trends 30° west of north compared to the 30° east of north of Judd 4, Tyrrell (1928, p. 200) proposed that it is the same intrusion, and also that it may be related to the quartz-porphyries to the south (Fig. 4). Kanaris-Sotiriou and Gibbs (1985) agree with these suggestions. The dyke is bordered on its eastern side by a thick (*c.* 8 m) felsite dyke and is cut roughly perpendicularly by a basic dyke (sampled at site 106) which, though it does not penetrate the felsite dyke, post-dates it, for the end of the cross-cutting dyke is chilled against the felsite. The composite dyke is also cut obliquely by another basic dyke 2 m thick (sampled at site 408).

Sites from the quartz-porphyry, near the centre (105, 107, 408C), adjacent to

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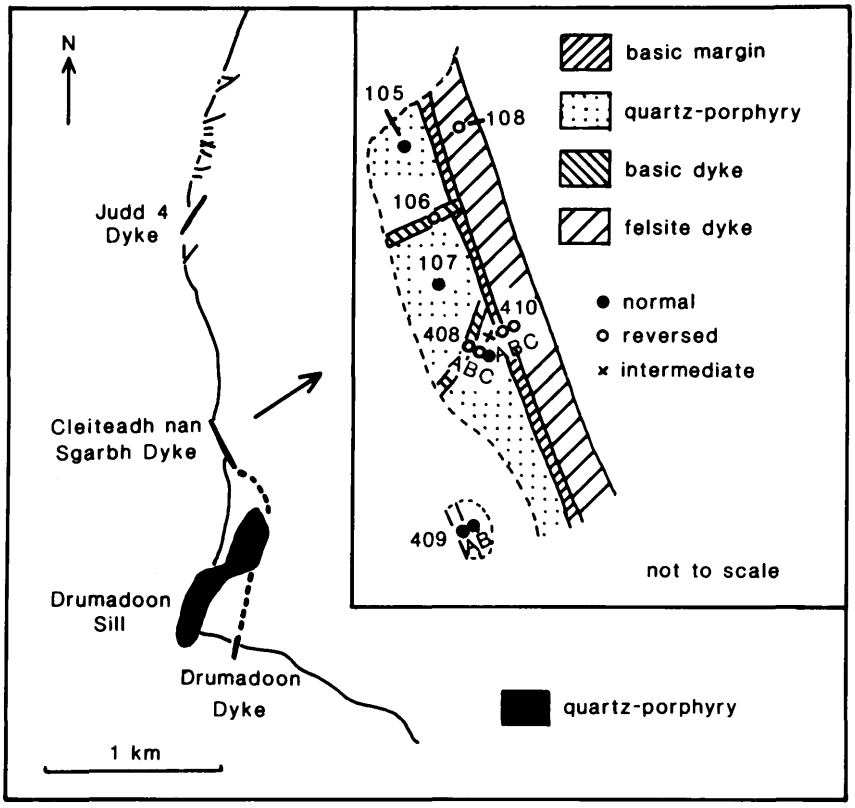


FIG. 4. A map showing dykes and sills, from the Tormore shore to Drumadoon Point (based on Tomkieiff 1969) with, inset, details of sampling sites and polarities in the Cleiteadh nan Sgarbh composite dyke.

the SW margin (409B), and from the SW dolerite margin (409A) are all normally magnetised but sites in the NE basic margin (283, 410B) have reversed polarity. However, as the thick felsite dyke (sites 108, 410C) is also R and was probably intruded after the composite dyke (Tyrrell 1928, p. 200; Kanaris-Sotiriou and Gibbs 1985) it probably overprinted the adjacent part of the composite dyke. Support for this can be found in the traverse from the felsite, through the basic margin, and into the quartz-porphyry (410): the felsite and margin are clearly R while the quartz-porphyry has an intermediate direction at site 410A-4, 0·56 m from the felsite, and normal directions at sites 410A-3 and -2, 0·6 and 0·7 m respectively, from the felsite.

A traverse from the basic dyke cutting obliquely into the quartz-porphyry (408) shows that the dyke and adjacent quartz-porphyry are reversely magnetised, but further away the quartz-porphyry is normal. The transition of polarity within the quartz-porphyry appears to be quite sharp.

All the other sites in this group of dykes are R.

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Thus the succession appears to be: intrusion of the normally magnetised composite intrusion, followed by the felsite dyke at a later time when the field was reversed and, finally, the cross-cutting basic dyke, 106, also in a reverse interval, possibly the same as that in which the felsite dyke was emplaced.

Other quartz-porphyries

The Drumadoon Sill (Fig. 4), a composite mass comprising a 30 m thick quartz-porphyry sandwiched between 1 m thick margins of tholeiite, was sampled in the central part (sites 221, 103) and upper (site 218) and lower margins (site 219). Nearby is the Drumadoon Dyke, solely quartz-porphyry, sampled at site 117, and almost certainly related to the Drumadoon sill and Cleiteadh nan Sgarbh Dyke (Tyrrell 1928, p. 200) (Fig. 4).

The Brown Head–Kilpatrick intrusion has poorly exposed contacts but consists of a mass of quartz-porphyry with a central band of felsite and is flanked on its eastern margin by another mass of felsite; there is no sign of any basalt or dolerite associated with it (Tyrrell 1928, p. 199). It was sampled by Dagley *et al.* (1978) (site Q003), and for this paper at Brown Head (sites 118, 246).

The Bennan intrusion is composite consisting mostly of quartz-porphyry, without basic margins at its westerly part where it is a sill but with dolerite margins where it steepens into a dyke at its eastern end. It was sampled in the quartz-porphyry by Dagley *et al.* (1978) (site Q004) and this paper (site 150), and in both basic margins (sites 151 and 200).

All these sites have normal polarity, with the exception that one of the three sites in the Brown Head–Kilpatrick intrusion (site 246) gave an intermediate direction.

Of the Arran quartz-porphyries listed by Tyrrell (1928 pp. 195–6) we have not sampled the following: the plug of Dun Dubh and nearby dykes, the dykes of the Sgiath Bhan ridge, of the Benlister and Monamore Glens and of the Allt Dhepin (the unsampled quartz-porphyry of Torr a'Bheannain is related to the Bennan Head intrusion (Herriot 1972)). No quartz-porphyry intrusions have been recorded in the north of Arran.

Radiometric data

A sample of quartz-porphyry from the Drumadoon Dyke was analysed using the ^{40}Ar–^{39}Ar step-heating method, following the procedure described in Mussett *et al.* (1980). Most of the step-ages fall in a narrow range and steps 9–15 satisfy the criteria of Lanphere and Dalrymple (1978), yielding a plateau age of $58{\cdot}5 \pm 0{\cdot}8$ (1σ) Ma (Fig. 5). (Other data are: total gas age 59 Ma; correlation plot gave an age of $58{\cdot}8 \pm 0{\cdot}9$ Ma, a $^{40}Ar/^{36}Ar$ contamination ratio of 286 ± 15 and an MSWD of 1·0; monitor was the Bern biotite 4B with an age of $17{\cdot}19 \pm 0{\cdot}12$ Ma, and the decay constants etc. recommended by Steiger and Jäger (1977) have been used).

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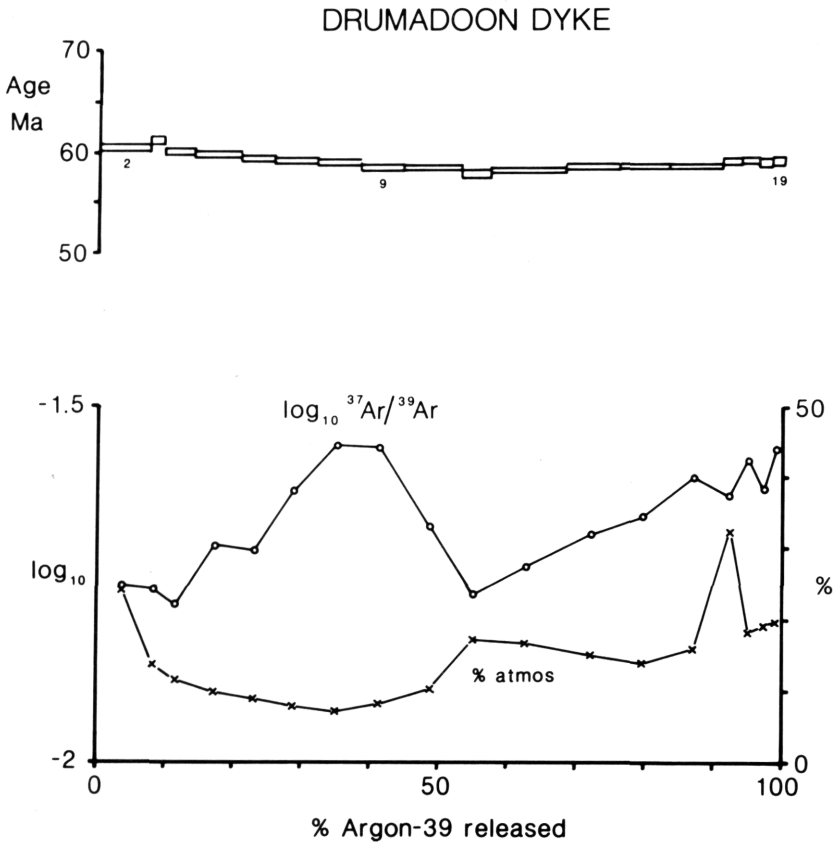


FIG. 5. Results of a ^{40}Ar–^{39}Ar analysis of the Drumadoon Dyke. Above is the age spectrum, while below are the $^{37}Ar/^{39}Ar$ ratio (proportional to the Ca/K ratio of the minerals releasing argon) and the percentage of atmospherically derived argon.

DISCUSSIONS AND CONCLUSIONS

Wherever a quartz-porphyry and, if present, the associated margin has been sampled it has had either normal magnetisation or there are good reasons to believe it has been overprinted by a later intrusion during a period of reverse magnetic polarity. Since the major quartz-porphyry bodies of Arran have been sampled it is proposed that all the Arran quartz-porphyries are normally magnetised.

Despite widespread sampling of the British Tertiary Igneous Province this is the first time that a correlation of polarity with a petrographic rock type has been found (though a correlation has been found with dyke trend in Fishnish, Mull (Dagley and Mussett 1978), and a partial correlation with position in northeastern Arran, where *c.* 70% of the dykes are normally magnetised, compared to only 17% for the sum of Arran dykes (Dagley *et al.* 1978). It is concluded that in Arran

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magmas of quartz-porphyry composition were available only for a limited period of time, while the magnetic field was normal. An examination of the polarity time-scale (Harland *et al.* 1982) at about 58·5 Ma ago, the age of the Drumadoon dyke, shows that the normal polarity period lasted no more than $\frac{1}{2}$ Ma. This provides an upper limit for the period of intrusion of quartz-porphyry magma.

Since the Arran Tertiary igneous rocks probably were intruded during a polarity succession R–N–R and the majority of rocks are R (Dagley *et al.*, 1978; Hodgson, 1981) it is suggested that normal polarity can provide a useful 'horizon' and represents an age of about 58·5 Ma.

Acknowledgments

We wish to thank Mr A. Herriot and Dr J. G. MacDonald for advice and encouragement. The dating analysis was performed by Mr M. Sankey and the work was made possible by NERC support: research grant GR3/4349, a grant for neutron irradiation and a studentship (to B.H.).

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MS. accepted for publication 16*th June* 1986